SMARTPID M5 PRO CUSTOM FIRMWARE

ProofPro Distilling Controller

User Manual SmartPID M5 PRO Custom Firmware

> Firmware: `proofpro`
> MQTT `schema_version`: `1`
> Draft date: 2026-05-29

This manual describes normal operation of the ProofPro custom firmware on the SmartPID M5 PRO hardware. Technical recovery, OEM firmware switching, installer builds, and boot-slot repair are covered separately in `docs/PROOFPRO_USER_MANUAL-technicalrecovery.md`.

Caution

- This controller is intended to control equipment under normal operating conditions. Controller failure, wiring error, software error, or misconfiguration may create unsafe output states.
- External limit controls, fuses, contactors, emergency shutoffs, and other independent safety devices must be installed and maintained for hazardous loads.
- Disconnect heaters, pumps, relays, contactors, and other hazardous loads before USB flashing, bench output testing, or recovery work.
- DC OUT 1 is connected to ESP32 GPIO12. GPIO12 is a boot strapping pin and can spike during USB auto-reset or ROM download entry. OTA is the normal update path for an installed ProofPro controller.
- Relay terminals may switch mains voltage depending on the installation. Do not probe or service AC terminals while energized unless using proper equipment and procedure.
- The installer is responsible for SSR and contactor ratings, heat sinking, wire sizing, fusing, grounding, strain relief, and enclosure safety.
- Firmware and MQTT control are not substitutes for independent safety cutoffs.

1. Specifications

Item	Specification
Hardware platform	M5Stack Basic/Gray-class ESP32, 16 MB flash
Firmware environment	`m5stack-core-esp32-16M-oem-layout`
Display	ILI9341 320x240 LCD
Buttons	Three mechanical buttons
Primary workflow	ProofPro Power screen / Power Direct control
Legacy concepts	Monitor, Standard, and Advanced are retained only as legacy/OEM-compatible concepts where supported
DC outputs	DC1, DC2
Relay outputs	RL1, RL2
Bench DC output rail	About 4.8 V carrier DC rail; measured high output about 4.6 V
Supported probe types	DS18B20, K-Type, PT100 2-wire, PT100 3-wire; NTC path present but not fully bench validated
Probe sample target	1 second
MQTT publish cadence	1 second default
MQTT topic root	`smartpidM5/proofpro/{topic_id}/`
Current schema	`schema_version: 1`

2. Front Panel And Power Screen

The controller boots directly to the ProofPro Power screen. This screen is the normal operating screen for local/manual control and for observing Proof remote control.

```
Figure 1. ProofPro Power screen layout
Placeholder: capture actual device screenshot and number the tiles below.
```

Screen callouts:

1. T1 temperature.

2. T2 temperature.
3. DC1 actual output power.
4. DC2 actual output power.
5. RL1 relay state.
6. RL2 relay state.
7. Remote state.
8. Reset.
9. Program status: `MAN`, `ACCEL`, `RUN`, or `END`.
10. Timer or end-condition display.

Display notes:

- The large value on a DC tile is actual driven output power.
- During acceleration, an element output shows acceleration power as the primary value. The queued run power may appear separately as smaller secondary text until acceleration ends.
- Relay tile ON/OFF follows the actual physical relay output.
- Managed relay armed state is distinct from physical output. In MQTT, `relay_engaged` means commanded or armed; `relay` means actual physical state.
- Disabled DC or relay tiles are darkened and skipped during selection.

Button guidance:

Button	Normal behavior
BtnA	Follow the left footer label; usually Up. In value editors, decrement.
BtnB	Select, menu, or confirm. Hold BtnB for Back where supported.
BtnC	Follow the right footer label; usually Down or Back. In value editors, increment.

Treat the on-screen footer labels as authoritative for each screen.

3. Wiring And Terminals

Use the manufacturer terminal markings and the wiring reference for the final installation. The table below summarizes the ProofPro output map confirmed on the bench unit.

Figure 2. ProofPro terminal/output map
Placeholder: terminal block photograph or diagram.

Terminal	GPIO	Function	Notes
DC OUT 1	GPIO12	DC1 PWM output	ESP32 strapping pin; keep low at reset
DC OUT 2	GPIO13	DC2 PWM output	DC output for Power Direct control
RL1	GPIO26	Relay 1	Dry-contact relay path / relay output
RL2	GPIO16	Relay 2	Dry-contact relay path / relay output

DC OUT voltage follows the board DC input rail. On the bench, the DC input was about 4.8 V and DC OUT high measured about 4.6 V. Use SSRs and interface devices that match the installed voltage and current requirements.

Figure 3. Example SSR wiring for DC OUT
Placeholder: DC OUT to SSR input, SSR output in heater circuit, independent fuse and safety cutoff shown.

Figure 4. Example acceleration relay/contactor wiring
Placeholder: RL contact driving a properly rated contactor or relay coil with appropriate protection.

Probe channel terminals expose:

```
GND / AIN1 / AIN0
```

Probe notes:

- PT100 3-wire was bench-confirmed with white lead to `GND` and red leads to `AIN1` and `AIN0`.
- PT100 2-wire T2 is bench-confirmed, but the red lead position matters. T1 2-wire terminal pairing still needs confirmation.
- DS18B20, K-Type, and PT100 3-wire have bench-confirmed reasonable readings.
- NTC support exists in firmware but remains unvalidated unless a later bench record says otherwise.
- Process temperatures below 0 C / 32 F or above 120 C / 248 F are treated as sensor errors. The UI should show `ERR`; telemetry reports `temp_valid:false`.

4. ProofPro Workflow

ProofPro uses the Power screen and Power Direct control as the normal operating workflow. Monitor, Standard, and Advanced are legacy/OEM-compatible concepts and should not be treated as normal operator workflows unless the current UI explicitly exposes them.

Power Direct provides:

- direct DC output duty control,
- acceleration phase,
- programmed run power,
- finish timer,
- finish temperature and source probe,
- relay modes,
- Remote-gated MQTT control from Proof.

5. Local Operation

The Power screen supports local/manual operation without Proof actively controlling outputs. Available operations depend on the selected tile and the current output mode.

5.1 Remote State

Remote controls whether MQTT output and program commands are accepted.

State	Meaning
`OFF`	Remote disabled. MQTT output/program commands are rejected or ignored.
`RDY`	Remote enabled. Proof can command the controller. Watchdog is not active yet.
`ON`	Active Proof session. Heartbeat is expected and watchdog protection is active.

Enable Remote from the Power screen before starting a Proof-controlled run. Disable Remote when the device should not accept remote output or program commands.

5.2 Status Labels

Label	Meaning
`MAN`	Manual/live output control. Program logic is not running.
`ACCEL`	Acceleration phase active. Element outputs use `accel_power`.
`RUN`	Programmed run phase after acceleration. Element outputs use run power.
`END`	Finish condition reached. Outputs are safe/off and latched until reset/start as configured.

5.3 Manual Power And Program Control

Use the DC tiles to adjust live output power. The value shown on the tile is the actual driven power after mode, acceleration, watchdog, and END latch effects.

When a program is running:

1. If acceleration is enabled, element DC outputs use `accel_power` until the acceleration threshold is reached.
2. After acceleration, element outputs use `post_accel_power`.
3. Auxiliary DC outputs are excluded from automatic acceleration and run power. They stay off unless explicitly commanded locally or remotely.
4. The finish timer starts when the selected runtime temperature reaches Timer Start Temp.
5. Finish by timer or finish temperature can place the controller in `END`.

Use Reset to clear `END` or watchdog safe state. Reset does not start a program by itself.

5.4 Relay Operation

Relay behavior depends on relay mode.

Mode	Local/operator meaning
`off`	Disabled and forced off.
`manual_on_off`	Local UI on/off only.
`acc_element`	Relay may run during acceleration when engaged.
`remote_other`	Proof directly controls the relay.
`cycle`	Firmware cycles the relay using ON time and cycle period.

Changing relay mode clears the relay command and turns the physical relay off. Local disengage of a managed relay is authoritative; Proof should not re-engage it unless the operator or app explicitly chooses to resume.

6. Proof / MQTT Remote Operation

Proof normally controls the device by sending program settings, enabling Remote, starting the run, and sending heartbeat messages while actively controlling.

Typical remote sequence:

1. Confirm retained `status` and `config`.
2. Enable Remote on the controller.
3. Send or confirm program settings.
4. Send `{"program_running":true}` to start.
5. Send `{"heartbeat":true}` about every 10 seconds while actively controlling.
6. Send `{"program_running":false}` to leave programmed ACCEL/RUN without a full safe/off shutdown, or `{"stop":true}` for full safe/off stop.

Remote command distinction:

Command	Result
`{"program_running":true}`	Starts programmed ProofPro power control.
`{"program_running":false}`	Returns to manual/live power control without forcing DC outputs to zero.
`{"stop":true}`	Forces DC outputs and relays off and clears program state.
`{"reset":true}`	Clears END/watchdog state; does not start a run.

Telemetry distinction:

Field	Meaning
`power`	Actual driven DC output percent after acceleration, watchdog, END latch, and mode effects.
`run_target_power`	Queued or target run percent after acceleration.
`relay`	Actual physical relay output state.
`relay_engaged`	Commanded or armed relay state for managed relay modes.

7. Parameter Settings

7.1 Program Parameters

ProofPro label	MQTT key	Range / values	Default / note
Accel Mode	`acc_mode`	`true` / `false`	Enables acceleration phase
Accel Temp	`accel_temp`	Temperature; `0` disables temperature transition	DSPR400 `dAST`
Accel Power	`accel_power`	0-100%	DSPR400 `dOUT`
Run Power	`post_accel_power`	0-100%	DSPR400 dial/run power
Timer Start Temp	`timer_start_temp`	Temperature	DSPR400 `dtSP`
Timer	`timer_s`	Seconds	Finish timer duration
Finish Temp	`finish_temp`	Temperature; `0` disables finish-by-temp	DSPR400 `dFSP`
Finish Temp Source	`finish_temp_source`	`CH1` or `CH2`	Selects probe for finish-by-temp
Finish Action	`finish_action`	`continue`, `end`, `shutoff`	`end` latches safe/off; `shutoff` is accepted as an alias for `end`

Program parameters are device-level. ProofPro has one finish temperature and one finish temperature source; there are no independent CH1/CH2 finish temperatures.

7.2 DC Output Modes

Mode	Meaning
`element`	Participates in acceleration and programmed run power.
`auxiliary`	Manual/direct output; excluded from automatic program power.
`off`	Disabled and forced off.

Accepted alias: `aux` maps to `auxiliary`. The misspelled legacy alias `auxilary` is also accepted by firmware.

7.3 Relay Modes

Mode	Meaning
`off`	Disabled and forced off.
`manual_on_off`	Local UI on/off only.
`acc_element`	Acceleration-managed relay.
`remote_other`	Proof directly controls the relay.
`cycle`	Firmware cycles the relay using `on_ms` and `cycle_ms`.

Accepted relay mode aliases:

Alias	Maps to
`remote`	`remote_other`
`reflux_timer`	`cycle`
`acc_sync`	`acc_element`
`manual`	`manual_on_off`
`on_off`	`manual_on_off`

7.4 Watchdog

Setting	MQTT key	Range / values
Watchdog Enabled	`watchdog_enabled`	`true` / `false`; disabling is rejected while Remote is enabled
Watchdog Timeout	`watchdog_s`	30-60 seconds

Watchdog behavior:

- Watchdog is active only during Remote `ON`.
- Enabling Remote forces watchdog enabled and clamps invalid timeout to default.
- Disable/unarm is accepted only when Remote is `OFF`.
- Proof sends heartbeat every 10 seconds while actively controlling.
- No heartbeat or accepted remote command traffic for `watchdog_s` seconds while Remote is `ON` trips watchdog safe.
- A trip forces DC1/DC2 to 0%, RL1/RL2 off, and returns Remote from `ON` to `RDY`.
- A later accepted command clears watchdog safe state and emits `watchdog_cleared`.

7.5 Clock / Timezone

Setting	MQTT command key / readback field	Notes
Timezone label	command/readback: `timezone_label`	IANA-style label such as `America/New_York`
POSIX timezone	command/readback: `timezone_posix`	ESP32 POSIX timezone string, not IANA
NTP enabled	readback/local setting: `ntp_enabled`	Retained config/local UI setting; not currently an MQTT command key
NTP host	readback/local setting: `ntp_host`	Retained config/local UI setting; not currently an MQTT command key
24-hour clock	command/readback: `clock_24h`	`true` for 24-hour display, `false` for 12-hour AM/PM

Clock commands are accepted regardless of Remote state because they do not energize outputs.

8. Application Examples

8.1 Alcohol Distilling With Acceleration

Example settings:

Setting	Value
Accel Mode	Enabled
Accel Temp	170 F
Accel Power	100%
Run Power	30%
Timer Start Temp	173 F
Timer	3:00:00
Finish Temp	200 F
Finish Source	CH1 or CH2, depending on probe installation
Finish Action	`end`

Operating sequence:

1. Confirm all wiring, probe placement, cooling, and independent safety devices.

2. Enable Remote if Proof will control the run.
3. Start the programmed run.
4. Element outputs configured as `element` heat at Accel Power.
5. At Accel Temp, status changes from `ACCEL` to `RUN`; element outputs drop to Run Power.
6. When the selected process temperature reaches Timer Start Temp, the finish timer starts.
7. The run ends when Finish Temp is reached by the selected source probe or when the timer expires.
8. Outputs go safe/off and `END` is shown.

```
Figure 5. Acceleration/run/finish timing diagram
Placeholder: show Accel Power to Run Power transition, timer start, and END.
```

8.2 Timer END Test

Example based on bench Run C:

Setting	Value
Accel Mode	Enabled
Accel Temp	155 F
Accel Power	100%
Run Power	35%
Timer Start Temp	155 F
Timer	60 seconds
Finish Action	`end`
Relay example	CH1 `acc_element`, CH2 `remote_other`

Expected result:

1. Program starts in `ACCEL`.
2. At 155 F, status changes to `RUN`.
3. Timer starts.
4. Timer expiration publishes `program_ended` with `reason:"finish_timer"`.
5. DC1/DC2 are 0%, RL1/RL2 are off, and Remote returns from `ON` to `RDY`.

8.3 Finish Temp Source CH1

Example based on bench Run D:

Setting	Value
Accel Mode	Disabled
Finish Temp	175 F
Finish Source	CH1
Timer	300 seconds
Finish Action	`end`

Only CH1 can trigger finish-by-temp. The expected END reason is `finish_temp`.

8.4 Finish Temp Source CH2

Example based on bench Run E:

Setting	Value
Accel Mode	Disabled
Finish Temp	190 F
Finish Source	CH2
Timer	300 seconds
Finish Action	`end`

Only CH2 can trigger finish-by-temp. The expected END reason is `finish_temp`.

8.5 Cycle Relay

Example based on bench Run F:

Setting	Value
Relay mode	`cycle`
ON time	`on_ms = 1000`
Cycle period	`cycle_ms = 5000`
Relay command	`CHx relay = true`

In cycle mode, `relay_engaged:true` means the cycle is armed. The physical `relay` field still alternates between ON and OFF according to the configured cycle timing.

8.6 Manual Regulator Mode

To use ProofPro like a manual power regulator:

1. Disable acceleration or leave the program stopped.
2. Set the desired DC output mode. Use `element` for the main output or `auxiliary` for a direct/manual output.
3. Adjust power from the Power screen or send live `CHx power` commands from Proof.
4. Watch actual `power` telemetry and the large DC tile value. Do not confuse retained Run Power with actual driven output.
5. Use `stop` or local controls to force outputs safe/off when finished.

9. Troubleshooting

Symptom	Likely cause	Action
Temperature shows `ERR`	Probe disconnected, wrong probe type, invalid route, or process temp outside 0 C..120 C	Check probe type, terminals, and wiring. For PT100 2-wire, try the other AIN terminal and retest.
Proof command does not affect output	Remote is `OFF`, output mode is `off`, or command is not accepted in current mode	Enable Remote, check retained config, and confirm live telemetry.
Remote shows `RDY` but not `ON`	Remote is enabled but Proof has not started an active session	Send heartbeat or a valid remote command.
Watchdog safe state	Proof heartbeat or accepted remote traffic stopped for longer than `watchdog_s`	Inspect MQTT connection, then send an accepted command or reset after confirming the process is safe.
Relay shows armed but not physically ON	Relay mode may be `cycle` or `acc_element`; `relay_engaged` is not the same as `relay`	Check both `relay_engaged` and `relay` telemetry.
Power tile shows 0 after stop while Run Power remains configured	`stop` forces actual output safe/off but does not erase saved program defaults	This is normal. Confirm `power`, not only `post_accel_power`.
DC OUT 1 energizes during USB flashing attempt	GPIO12 boot strapping / USB auto-reset hazard	Disconnect hazardous loads. Use OTA for normal updates. See the technical recovery manual for bench recovery.
Output state is uncertain	Need output readback	Send `{"diagnostics":"outputs"}` or use the serial `diag` command on the bench.

10. Commissioning And OTA

10.1 First Boot Captive Portal

On first boot with no saved WiFi credentials, the controller starts an access point:

```
SmartPID-XXXXXX
```

Connect a phone or computer to that network. If the captive portal does not open automatically, browse to:

```
http://192.168.4.1/
```

Enter WiFi SSID/password and MQTT host, port, username, and password. The device saves the settings and reboots into client mode.

To force the portal later, hold BtnA during boot.

10.2 Verify MQTT

Subscribe to retained status:

```
mosquitto_sub -h <broker-ip> -u <user> -P <password> \
  -t 'smartpidM5/proofpro/+/status' -v
```

Expected status shape:

```
{
  "serial": "000C3BA7C0E8FC",
  "firmware": "proofpro",
  "schema_version": 1,
  "unit": "F",
  "remote_enabled": true,
  "remote_state": "RDY",
  "watchdog_enabled": true,
  "watchdog_s": 30
}
```

Retained `status` is the discovery/onboarding source of truth. Retained `config` contains editable/default program, DC output, clock, and relay settings.

10.3 OTA Updates

OTA is the normal update path when ProofPro is running and connected to WiFi:

```
pio run -t upload --upload-port <device-ip>
```

Because the default environment is `m5stack-core-esp32-16M-oem-layout`, this command builds and uploads the correct hardware image for converted devices.

Do not USB-flash with heaters or other hazardous loads connected to DC1/DC2/RL1/RL2. USB flashing is a bench recovery path only.

Appendix A. ProofPro MQTT Operator/API Reference

This appendix is a shortened operator/API reference. The canonical and complete schema is `docs/MQTT_SCHEMA.md`. Migration, firmware restore, firmware switching, installer, and USB recovery commands are intentionally omitted from this manual.

A.1 Topic Root

```
smartpidM5/proofpro/{topic_id}/
```

Current bench topic ID:

```
791402d5ac0fe1
```

A.2 Published Topics

Topic suffix	Retained	Purpose
`status`	Yes	Device identity, unit, firmware/schema, Remote state, watchdog settings
`config`	Yes	Program defaults, DC output modes, clock readback, relay modes/timing
`power/CH1`	No	CH1 power-mode telemetry
`power/CH2`	No	CH2 power-mode telemetry
`events/standard`	No	Device and channel events

Legacy `dynamic/CH1`, `dynamic/CH2`, and `events/advanced` may exist for compatibility. ProofPro custom workflow should prefer `power/CHx`.

A.3 Subscribed Topics

Topic suffix	Purpose
`commands`	JSON command dispatch
`profiles/update/#`	Legacy profile compatibility; not part of normal ProofPro workflow

A.4 Command Examples

Request status/config refresh:

```
{"status": true}
```

Heartbeat:

```
{"heartbeat": true}
```

Start programmed run:

```
{"program_running": true}
```

Leave programmed run without full safe/off stop:

```
{"program_running": false}
```

Full safe/off stop:

```
{"stop": true}
```

Reset END/watchdog state:

```
{"reset": true}
```

Program settings:

```
{
  "acc_mode": true,
  "accel_temp": 170,
  "accel_power": 100,
  "post_accel_power": 35,
  "timer_start_temp": 170,
  "timer_s": 3600,
  "finish_temp": 200,
  "finish_temp_source": "CH1",
  "finish_action": "end"
}
```

DC output commands:

```
{"DC1 dc_mode": "element", "DC2 dc_mode": "auxiliary"}
```

```json
{"CH1 power": 35}
```

Relay mode and command:

```json
{"CH1 relay_mode": "cycle", "CH1 on_ms": 1000, "CH1 cycle_ms": 5000}
```

```json
{"CH1 relay": true}
```

Watchdog:

```json
{"watchdog_enabled": true, "watchdog_s": 30}
```

Clock/timezone:

```json
{
  "timezone_label": "America/New_York",
  "timezone_posix": "EST5EDT,M3.2.0,M11.1.0",
  "clock_24h": false
}
```

Output diagnostics:

```json
{"diagnostics": "outputs"}
```

Test chirp:

```json
{"chirp": true}
```

Program END plays three 2670 Hz tones, 3 seconds each, with 1 second gaps. Explicit safe/off stop plays three 1800 Hz tones, 0.5 seconds each, with 0.5 second gaps.

A.5 Field Semantics

Field	Meaning
`remote_state`	`OFF`, `RDY`, or `ON`; see Section 5.1.
`program_running`	`true` when a channel is under ProofPro ACCEL/RUN/END logic; `false` in manual/live mode.
`power`	Actual driven DC output percent.
`run_target_power`	Queued/target run percent after acceleration.
`relay`	Actual physical relay output.
`relay_engaged`	Commanded/armed relay state.
`ended`	Device/program END state reflected in telemetry.
`latched`	Outputs are latched safe/off until reset/start.
`timer_remaining_s`	Remaining finish timer seconds; freezes if END occurs early.
`timer_frozen`	Timer display has frozen because END occurred before timer expiry.

A.6 Remote Gating

Accepted regardless of Remote:

- `status`
- `heartbeat`
- `stop`
- `pause`
- `resume`
- `reset`
- `chirp` / `audio:"chirp"`

Require Remote enabled:

- `program_running`
- `start`
- output power commands
- relay commands
- program parameter writes
- DC and relay mode writes

Appendix B. Proof Integration Audit

This appendix is for integrators. Omit it from operator-only distributions if Proof database audit behavior is not relevant.

Proof audit records are Proof-side database records, not firmware MQTT payload fields. Firmware does not currently require `origin` or `trigger` fields in MQTT commands.

For every outbound MQTT command Proof sends to `commands`, Proof should persist:

- timestamp,
- run ID when associated with a run,
- device ID,
- topic,
- payload,
- origin,
- trigger.

Recommended `origin` values:

Origin	Meaning
`operator`	Direct operator action in Proof UI
`automation`	Proof automation or run logic
`restore`	Reconnect/reboot restore flow
`safety`	Proof-side safety action
`heartbeat`	Periodic active-session heartbeat
`system`	System/settings synchronization or service action

Recommended `trigger` examples:

Trigger	Typical command source
`run_start`	Operator starts a ProofPro run
`manual_stop_button`	Operator presses stop
`relay_toggle_button`	Operator toggles RL1/RL2
`reconnect_restore`	Proof restores an active run after reconnect
`heartbeat_tick`	Proof heartbeat loop
`settings_sync`	Proof writes synchronized settings
`program_end_cleanup`	Proof cleanup after firmware END

When an outbound command is tied to a run, Proof should also write a run event with type `mqtt_command_sent`.

When Proof receives a firmware `program_ended` event, Proof should persist a decision event with type `proofpro_program_end_decision`. Preserve an optional future firmware `source` field as `firmware_source` when present. Do not require `source`; current firmware may omit it.

Appendix C. DSPR400 Vocabulary Cross-Reference

DSPR400 concept	ProofPro setting / field	Notes
Distilling power / dial power, for example `P30`	Run Power / `post_accel_power`	Element output after acceleration
`dAST`	Accel Temp / `accel_temp`	Acceleration end temperature
`dOUT`	Accel Power / `accel_power`	Element output during acceleration
`dtSP`	Timer Start Temp / `timer_start_temp`	Temperature that starts finish timer
`dt`	Timer / `timer_s`	Finish timer duration
`dFSP`	Finish Temp / `finish_temp`	Device-level finish threshold
Finish source	Finish Temp Source / `finish_temp_source`	`CH1` or `CH2`
`dEO` / finish behavior	Finish Action / `finish_action`	`continue`, `end`, or `shutoff` alias
Main element	`dc_outputs.DCx.mode:"element"`	Uses accel/run program power
Auxiliary element	`dc_outputs.DCx.mode:"auxiliary"`	Manual/direct output
Auxiliary acceleration relay	Relay mode `acc_element`	May be operator-disengaged
Reflux/cycle relay	Relay mode `cycle`	Uses `on_ms` and `cycle_ms`
Remote relay/manual test	Relay mode `remote_other`	Proof direct relay control
END condition	`program_ended.reason`	`finish_timer`, `finish_temp`, or `finish`

Appendix D. Validation Status

This appendix reflects the current documentation and bench status. Do not treat unvalidated items as production-confirmed behavior.

Area	Current status
Build target	`m5stack-core-esp32-16M-oem-layout` is the current default hardware build
OTA	Bench-confirmed to `10.0.1.60`
DC1/DC2/RL1/RL2 GPIO map	Confirmed
Normal boot/OTA output state	Normal firmware boot and OTA path hold outputs safe/off
USB flashing hazard	Confirmed DC1/GPIO12 spike during USB auto-reset/download entry
Power screen	Boots directly to Power screen; T1/T2, DC1/DC2, RL1/RL2, Remote, status, timer shown
Accel transition	Bench-confirmed; device-wide ACCEL to RUN transition
Timer END	Bench-confirmed
END-before-timer freeze	Cleared; remaining time freezes when END occurs early
DS18B20	Bench-confirmed reasonable readings
K-Type	Bench-confirmed after ADS1119 route work
PT100 3-wire	Bench-confirmed with documented calibration offsets
PT100 2-wire	T2 route confirmed; T1 still needs confirmation
NTC	Code path present; not currently bench validated
Watchdog	Implemented; still requires focused bench validation unless later records clear it
Cycle relay	Implemented; still requires focused bench validation unless later records clear it
Proof remote relay regression	Needs final hardware retest with current firmware unless later records clear it
Run-ending audio	No onboard microphone; final DSPR400-style tone sequence still depends on recording/timing extraction